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Analysis of the impact of artificial intelligence on electricity consumption

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***Abstract*—The in-depth development and widespread application of artificial intelligence technology will have significant impact on electricity demand. However, there are few comprehensive analyses. This article will analyze the direct and indirect impact respectively. For direct impact, artificial intelligence requires various data centers to provide huge computing and storage capabilities, which will increase the electricity demand of data centers; for indirect impact, artificial intelligence will also optimize the time and space distribution of electricity demand and deepen energy conservation and power saving in the whole society. Research shows that the development of artificial intelligence will increase the electricity consumption of data centers, and it is necessary to plan ahead for the infrastructure related to the electricity consumption of artificial intelligence; artificial intelligence plays an important role in promoting the development of demand response, and a large-scale joint dispatch mechanism of electricity and computing power for the power market should be explored; data centers have a large space for energy conservation and carbon reduction, and a sound electricity price mechanism that comprehensively considers green electricity consumption and energy efficiency should be established to promote data centers to widely participate in green electricity transactions and actively reduce their own PUE.**

Keywords-artificial intelligence; electricity demand; demand response; energy efficiency

I. INTRODUCTION

The in-depth development and widespread application of artificial intelligence technology will have an important impact on electricity demand. However, there are currently few comprehensive analyses.

There are many studies discussing the development of artificial intelligence [1-2]. The journey of artificial intelligence began in the 1950s, with milestones including Alan Turing's "Turing Test" and the Dartmouth Conference, which coined the term "artificial intelligence". In recent years, artificial intelligence has entered a stage of rapid development and widespread application, gradually transforming from early theoretical and laboratory research to large-scale commercial practice. In the 1980s, David Rumelhart and others introduced the backpropagation algorithm, which enabled the creation of multi-layer neural networks. This marked the beginning of machine learning and the ability to train networks with thousands of parameters and multiple layers. In 2017, researchers described how to train very large neural networks, the so-called Transformers. This opened the era of generative artificial intelligence and large language models, which contain billions to trillions of parameters and are trained on huge data sets. The performance of these models has shocked researchers and the public, and they are widely used by millions of people every day.

There are also some studies on the energy consumption of artificial intelligence models [3-5]. Some scholars have conducted calculations and analyses on typical AI models, and demonstrated the future evolution trend of AI model energy consumption through comparison. But the above research is not comprehensive, where the analyses on AI promoting demand response and energy conservation are not included.

This article will analyze it from two aspects: direct impact and indirect impact. From the perspective of direct impact, artificial intelligence requires various data centers to provide huge computing and storage capabilities, which will directly increase the electricity demand of data centers; from the perspective of indirect impact, artificial intelligence will also optimize the spatial and temporal distribution of electricity demand and deepen energy conservation and power saving in the whole society.

II. THE DIRECT IMPACT OF ARTIFICIAL INTELLIGENCE ON ELECTRICITY

A. Data Center Overview

A data center is a large-scale data storage and processing facility used to store, process, manage and distribute Internet-related data and applications. It is usually composed of a series of network equipment, servers, storage equipment, backup equipment, security equipment, refrigeration equipment, power supply equipment, etc. They are usually operated and managed by large Internet companies, telecom operators, cloud computing service providers, government agencies, etc.

Data storage and management: Data centers provide data storage and management services to store and manage various types of data, including documents, images, videos, audio, etc.

Server hosting and management: The data center provides server hosting and management services, manages and maintains a large number of servers, and provides users with server leasing, maintenance, upgrade and other services.

Network operation and management: Data centers provide network operation and management services, providing users with bandwidth, network access, network monitoring, traffic control and other services.

Security assurance and management: Data centers provide security assurance and management services to ensure the security and privacy of user data, including physical security, network security, data backup, etc.

Cloud computing services: More and more data center providers provide cloud computing services, providing virtualization, elastic computing, cloud storage, cloud database and other services. How much proportion of AI is implemented on cloud will be discussed in further research.

B. *Power Consumption in Data Centers*

The electricity consumption of data centers can usually be divided into two parts, as shown in Table I: IT equipment electricity consumption and infrastructure electricity consumption. Among them, IT equipment mainly includes servers, storage devices, network equipment, etc., and infrastructure mainly includes cooling systems, power distribution devices, etc. The power consumption structure of a typical data center is given in Figure 1.

TABLE I. POWER CONSUMPTION STRUCTURE OF DATA CENTERS

Electricity usage category	Specific equipment
IT Equipment Power Consumption	Servers: including GPUs, CPUs, and memory that analyze and process data, are the largest source of energy consumption in data centers.
	Storage devices: including hard disks (HDDs) and solid-state drives (SSDs), etc., have relatively low unit power consumption, but large-scale storage capacity will generate large energy consumption.
	Network equipment: including switches, routers, and other communication equipment used for data transmission and interaction, which account for a relatively small proportion of energy consumption.
Infrastructure electricity	Cooling system: used to maintain the appropriate temperature of the equipment in the data center, it is the second largest source of energy consumption in the data center. The energy demand is related to cooling technology and climate conditions.
	Other facilities: including uninterruptible power supply (UPS), power distribution facilities, lighting and security systems, etc.

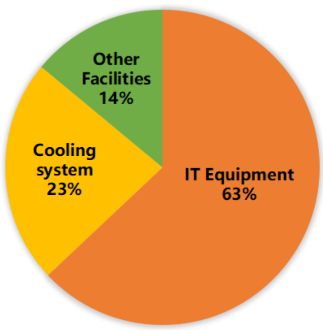

Figure 1. Typical data center (PUE=1.6) power consumption structure

PUE (power usage efficiency) is usually used to measure the energy composition of data centers, that is, the ratio of total energy consumption to IT equipment energy consumption (PUE = total energy consumption of data centers / IT equipment energy consumption). Ideally, PUE is close to 1.0, that is, most of the energy is used for IT equipment operations. From the perspective of electricity consumption, taking a typical data center with PUE=1.6 as an example, its IT equipment electricity consumption accounts for nearly 2/3, and the cooling system electricity consumption accounts for about 1/4.

C. *Factors affecting data center electricity consumption*

One is the complexity of the calculation model. On the one hand, complex computing models, such as deep learning, require a large amount of parallel computing power and rely on GPUs or other dedicated hardware. The power of these hardware itself is greater than that of traditional CPUs. When a large number of such hardware is deployed in the data center, it will consume a lot of electricity.

On the other hand, larger models have more parameters, and the corresponding training and inference processes require more calculations; complex models also require more data for training, and processing the data also consumes more power. The energy consumption of artificial intelligence models is shown in Figure 2.

Taking ChatGPT as an example, when upgrading from ChatGPT-3 to ChatGPT-4, its parameters increased from 175 billion to 1.76 trillion, an increase of 906%, and the size of the training data set increased from 300 billion marks to 13 trillion marks, an increase of 4233%, correspondingly. The training electricity consumption increased from 1.29 million kilowatt hours to 50 million kilowatt hours, an increase of 3776% [6]. As artificial intelligence technology continues to make breakthroughs, the complexity of future computing models will increase significantly, and the demand for intelligent computing in data center tasks will also increase significantly.

The second is the scale of user call demand. Once the computing model is trained, although less energy is required to run the model in a single run to respond to user queries or perform tasks, the energy consumption will also be huge if user calls are in high demand.

Take Google, the world's largest search engine, as an example. It currently responds to about 9 billion searches every day. If ChatGPT's generative artificial intelligence is integrated on a large scale in all Google searches, and the power consumed by a ChatGPT query is 0.0029 degrees, then An additional 260 million kilowatt-hours of electricity will be consumed every day, and an additional 95.3 billion kilowatt-hours will be consumed every year (approximately 1.0% of China's total social electricity consumption in 2023) [7]. As general and specialized artificial intelligence is rapidly promoted and applied to all walks of life, its high-frequency and large-scale calls to different computing models will be extremely large, resulting in huge demand for electricity.

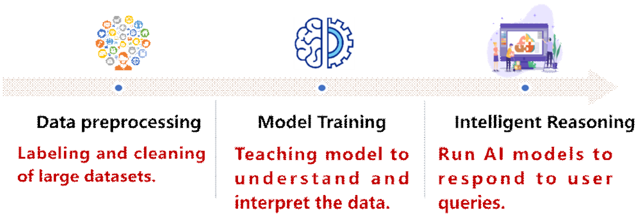

Figure 2. Energy consumption of artificial intelligence models

The third is data center energy efficiency. In order to ensure the stable operation of IT equipment, the data center needs to be cooled by refrigeration equipment such as air conditioners and chillers, which consumes a lot of electricity. This part of the energy consumption is affected by both equipment factors and environmental factors, including the heating conditions of various equipment, the operating conditions of the refrigeration system, the thermal conductivity of the building envelope, the local climate environment, etc.

It is usually reflected by the above-mentioned PUE value. Lowering the PUE value can significantly To reduce cooling energy consumption. For example, in 2022, the lowest PUE of data centers in the United States is 1.08, and the lowest PUE of data centers in my country is 1.09, which is only 0.9% higher than that in the United States. However, with the same IT equipment power consumption, my country's cooling power consumption is 12.5% more [8]. In order to reduce construction and operating costs and optimize the operating environment of IT equipment, data center operators have great motivation to improve energy efficiency, and PUE will continue to decline.

D. Data Center Power Consumption Trends

Since 2010, despite strong growth in demand for data center services, data center energy use (excluding encryption) has grown only modestly, in part due to improvements in IT hardware and cooling efficiency, and a shift from small, inefficient enterprise data centers to more efficient cloud and hyperscale data centers.

Despite significant efficiency gains, the rapid growth in the workload handled by large data centers has caused energy use in this area to increase significantly over the past few years, increasing by 20-40% per year. Between 2017 and 2021, the combined electricity consumption of Amazon, Microsoft, Google, and Meta more than doubled, increasing to approximately 72 TWh in 2021[9].

In the future, as technology continues to advance, artificial intelligence will better understand and predict user needs, provide more personalized, efficient, and intelligent services, and profoundly impact and change all walks of life. In terms of energy and power, it optimizes power distribution and dispatching, integrates the management of renewable energy and load resources, and carries out advanced fault detection and preventive maintenance. In terms of manufacturing and warehousing, it realizes the automation and intelligence of the production process and optimizes inventory management. In terms of transportation and logistics, it improves the safety and reliability of self-driving cars and optimizes traffic flow and logistics management. In terms of medical health, it accelerates the development of new drugs, diagnoses diseases and assesses risks more accurately, and provides personalized medical care. In terms of education and training, it provides customized teaching and tutoring services, assists teachers in managing classrooms and students, and establishes virtual laboratories. In terms of financial services, it provides personalized financial consulting and product recommendations, and promptly detects and prevents fraud. In terms of environmental protection, it analyzes climate change trends, optimizes the management and utilization of natural resources, and monitors environmental pollution.

According to the International Energy Agency (IEA), in 2022, global data centers will consume about 460 terawatt hours of electricity (equivalent to 2% of the global total demand). As generative artificial intelligence (AI), which requires a lot of computing to run, becomes more and more popular, the global data center power consumption in 2026 will be up to 2.3 times that of 2022, and may expand to 620 to 1050 terawatt hours.

In terms of regions [10], for data centers that require short latency for computing tasks, their distribution is mainly around economically developed core cities, so as to be close to users; for other data centers, the layout should be based on comprehensive consideration of latency, electricity costs, climate conditions, etc. In the context of the "East Data West Computing" project, the eastern region will leave the business with high latency requirements and transmit the data with low latency requirements to the western region for computing, so as to fully enjoy the abundant local water, electricity and other resources and reduce costs.

III. Indirect Impact of AI on Electricity

A. Promoting demand response

Mainly by promoting users to participate in demand response and data centers to directly participate in demand response, including: empowering energy-consuming equipment, automatically tracking changes in grid frequency, voltage and energy prices, automatically organizing energy consumption behavior, and participating in demand response directly or through load aggregators. The data center itself supports the peak-shaving and valley-filling of the power system by adjusting the time and location of the calculation load execution. Typical load curve of data center is shown in Figure 3.

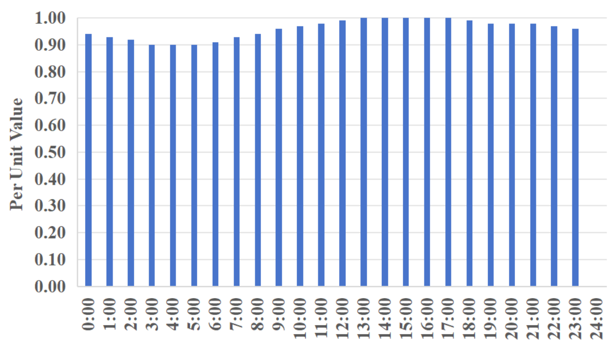

Figure 3. Typical load curve of data center

Taking the maximum load day of a certain city(The following is referred to by ZH) in summer 2023 as an example, when supply and demand are tight, if the data center load at the time of ZH's maximum load can be transferred to other locations, ZH's maximum load can be reduced by 17%; if the peak-to-valley difference of the data center on that day is used as the adjustment amount, by peak shaving or valley filling, ZH's maximum load can be reduced by 1.8% or the minimum load can be increased by 2.7%.

B. Deepen energy conservation and electricity saving in the whole society

Empowering energy efficiency services. For example, in the construction field, AI-driven building management systems can track occupancy, weather, usage patterns, etc., minimizing lighting, heating and cooling energy consumption. Taking Google's data center energy saving as an example, the Deep Mind model is used to predict the temperature and pressure of the data center in the next hour, and energy-saving control is performed with server temperature as the control target, reducing cooling energy consumption by 40% compared with the traditional manual control mode.

Data center itself: PUE is a key indicator for measuring energy efficiency [11-12]. According to the Uptime Institute's 2022 survey data, the current average PUE of large-scale data centers in the world is as high as 1.55, while most new data centers can reach PUE of about 1.3-1.25. In the future, with the further tightening of policies and the further development of technology, more and more advanced energy-saving technologies will be more widely used in data centers, driving the further decline of PUE. It is expected that by 2027, PUE will enter the 1.0x era. In large data center parks, heat recovery, as a new energy-saving solution, has gradually been implemented and successfully practiced. With the increasing attention paid to policies, more and more data centers are paying attention to and attaching importance to the value of recycling waste heat. Through heat channels, excess heat from data center equipment is directed to other facilities, improving waste heat utilization and saving energy costs.

On the one hand, the application of advanced liquid cooling technology can effectively reduce the power consumption of the cooling system. For ZH, under the same computing task, due to differences in temperature, the summer load is 12.6% higher than the winter load, and the energy-saving potential is considerable. On the other hand, the improvement of chip efficiency and algorithm efficiency will also slow down the growth rate of data center energy consumption. According to relevant research in Computer magazine, Google developed the AI model GLaM about 18 months later than ChatGPT-3. Even though GLaM has 6 times more parameters than the latter and is significantly more complex, its training power consumption is only 35% of the latter [3].

V. Conclusions

(1) The development of artificial intelligence will increase the electricity consumption of data centers. It is necessary to plan ahead for infrastructure related to the electricity consumption of artificial intelligence. Electricity planning at all levels should be closely linked to the corresponding artificial intelligence development plan, and sufficient and stable electricity infrastructure should be planned in advance to ensure the electricity demand for artificial intelligence. (This will be discussed in further research.)

(2) Artificial intelligence plays an important role in promoting the development of demand response. We should explore a large-scale joint dispatching mechanism of power and computing power for the power market, build a model-data fusion-driven grid coordination control mode based on demand and problem orientation, make full use of the characteristics of data centers that can dynamically adjust work tasks and power loads in time and space, and consider the adjustment potential of UPS and other power storage equipment configured in data centers, form a comprehensive demand response capability, and promote the coordinated interaction of source, grid, load and storage.

(3) Data centers have a large space for energy saving and carbon reduction. We should establish and improve an electricity price mechanism that comprehensively considers green electricity consumption and energy efficiency, so as to promote data centers to widely participate in green electricity transactions and actively reduce their own PUE. Relevant market players should do a good job in user docking, improve service capabilities, and innovate business models.

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